Project Report

End-to-End Sales Forecasting & Demand Intelligence System

Prepared for: Head of Supply Chain & Chief Financial Officer (CFO)

Executive Summary

This report summarizes findings from an analysis of the company's historical sales data and a three-month sales forecast. We examined sales trends, seasonality, unusual sales events, and product demand patterns to support inventory and financial planning. Three forecasting models were compared; XGBoost was selected for operational use due to its lowest MAE and MAPE, though SARIMA produced a slightly lower RMSE and is used to express forecast confidence ranges. The forecast points to moderate fluctuations over the next three months, reinforcing the need for flexible inventory planning. Segmentation analysis further identified distinct stocking strategies for high- and low-demand products. Overall, this system can reduce stock shortages, lower holding costs, and strengthen financial planning.

1. Key Findings

Sales show a clear long-term upward trend with strong seasonality: demand peaks during festive and holiday periods, especially toward year-end, and fluctuates monthly with promotions and seasonal demand.

Model	MAE	RMSE	MAPE
SARIMA	19,244.49	19,950.07	20.53%
XGBoost	17,909.13	20,357.71	18.13%
Prophet	20,250.79	22,318.41	21.86%

XGBoost achieved the lowest MAE and MAPE and is the preferred model; SARIMA's lowest RMSE makes it useful for confidence intervals.

2. Three-Month Sales Forecast

Month	XGBoost Forecast	SARIMA 95% Range
Month 1	44,274	16,993 – 76,572
Month 2	23,885	9,596 – 70,975
Month 3	54,557	41,319 – 103,149

Actual results may vary within these ranges due to promotions, holidays, and shifting customer demand.

3. Top Three Anomalies Detected

Period	Pattern	Likely Cause
November	Unusually high sales	Black Friday and festive promotions
January	Unusually low sales	Post-holiday slowdown; inventory replenishment
December	Unusually high sales	Christmas / year-end shopping and seasonal discounts

Both Isolation Forest and Z-score methods confirmed these periods, distinguishing normal seasonal demand from exceptional events.

4. Product Demand Segmentation & Stocking Strategy

Segment	Characteristics	Recommendation
High Volume, Stable	Consistently high, predictable demand	Maintain high stock; automatic replenishment
Growing Demand	Rapid increase, strong potential	Increase stock gradually; monitor monthly
Low Volume, Volatile	Irregular demand, higher risk	Keep limited stock; reorder as demand rises
Declining Demand	Falling sales, low interest	Reduce inventory; discount to clear stock

5. Business Recommendations

1. Use XGBoost for operational forecasting — it delivers the lowest MAE (17,909.13) and MAPE (18.13%), enabling more accurate inventory planning and fewer stock shortages or surpluses.

2. Increase inventory ahead of peak periods — historical data and anomaly detection confirm sales spikes in November and December, so building stock beforehand prevents stockouts and captures peak-season revenue.

3. Apply demand-based inventory management — clustering identified stable, growing, volatile, and declining segments, allowing the business to cut holding costs and improve turnover through segment-specific stocking.

6. Risk & Limitation

This system relies primarily on historical sales data. Unexpected events — economic shifts, supply chain disruptions, competitor pricing, regulation, or major promotions — can materially affect future sales before the model adapts. Forecasts should be refreshed regularly with new data and paired with business judgment before major decisions. Overall, combining forecasting, anomaly detection, and segmentation gives the business a stronger, data-driven basis for inventory and financial planning.